

PROCESS

This project started with the question "How can I use material already available at the Smissenbroek farm to make heating Isabelle's poorly insulated home easier?".

The research became not purely material, but a historical deep-dive into how we used to build en heat homes. How our perception of thermal comfort has changed, and how that might even be affecting our health. I talked to Robin Pender, a researcher and conservator of historical homes. Thermal comfort did not have much to do with the air temperature, and much more with radiant heat and heat loss.

My insulation material would be focused on that radiant heat loss; being placed on the outside of the walls, around and next to places of rest.

I used Narrowleaf Cattail (*Typha angustifolia*), an unused plant with a fiber that already provided excellent insulating and hydrophobic properties. It was important to me that the fibres stayed visible, and weren't hidden away.

Since the fibres were very short, a technique like wet felting wasn't enough to keep them stable. So, after some experimenting I opted for temporarily stabilising the fibre with Soluvlies and quilting the fibres together in 1x1 cm squares. The quilt stitching was also done with a cotton thread to ensure the biodegradability of the material.

Further aesthetical research was done by successfully dyeing the cattail with natural dyes (Madder and purple cabbage), and applying a pattern via textile ink and silk-screening.